

The Courage to Teach Nursing Innovation

Innovation Education, Content, and Strategies

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Innovation is not typically learned in many Master of Science in Nursing programs. This article discusses the design of a required 3-credit digital innovation course guided by the American Association of Colleges of Nurses (AACN) Essential domains focusing on nursing practice, planning care, and optimizing system effectiveness.¹ For students in this course, health care innovation is described as any new idea, nursing practice, model of care, clinical measures, or health care technology focused on value to health and health care; the innovation can be a product, service, or process.

Curriculum

The curriculum followed the school's Nursing Healthcare Innovation Impact Program (HiiP) Innovator Pathway (IP), daring the student to innovate using 4 phases. Phase 1 focuses on idea generation; the student identifies the core idea and its critical essential elements. The student connects the idea with a specific problem or opportunity and continues to iterate on creative solutions. Phase 2 focuses on idea development and refinement by conducting customer discovery, engaging in competitive benchmarking, developing detailed concepts of the idea, assessing the feasibility of the innovation, and developing a model (prototype). Phase 3 focuses on developing a business case and return on investment, identifying investment funds, producing the innovation, and developing a marketing plan. Phase 4 focuses on launching the innovation to the market, generating awareness of and creating interest in the innovation, and developing metrics for success.

Course

The project-based course taught students how to produce and successfully lead nursing innovation. The course

begins with the student understanding the culture of innovation and how their leadership style can impact the commitment of staff to identify solutions to problems. Next, the students worked in small groups completing idea generation, with the outcome being an innovation they would design throughout the class. The innovation development and refinement topics included conceptual frameworks, the Theory of Inventive Problem Solving principles that focus on patents, human factor engineering, regulation standards and guidelines (eg, FDA, risk management), and identifying patents that are like the group innovation project. The course focused on building skills for launching, promoting, and monitoring the innovation, including reviewing implementation science and evidence-based practice models and how they can be used with innovation implementation. Technology implementation models and how these models can be used with designing, users/stakeholders' requirements, deployment, and evaluation of the innovation were included in the course.

Teaching Strategies

Teaching involved multiple methods, beginning with students reviewing the literature and presenting innovative topic findings. Guest lectures engaged the students by connecting the class content with practicing professionals highlighting real-world applications. These guests did not charge for their services. They included a nurse innovator, a chief executive officer, and an electrical engineer of a medical innovation company, the HiiP director, a risk manager, and an implementation scientist. Students commented that the guest speakers helped them learn innovative content. Speakers addressed leadership qualities, teamwork, standards, marketing, return on investment, fundraising, and many more drivers of innovation. Video-based talks from YouTube were integrated into the course content, further engaging the student by viewing content versus reading articles. A student reported that she "liked how the professor arranged the modules and also embedded clips of videos of her explaining some of the concepts." The digital design engineers assisted with developing simulations to support further learning. Students indicated appreciating the mix of videos, readings, and interactive tasks. Students stated that the mixed course content kept their interest, allowed them

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to remain engaged, and enabled them to understand innovation and its essential concepts.

Assignments

Assignments were created to promote active learning by engaging students through the HiiP innovation pathway. In their first assignment, the students used video technology to introduce themselves to the class and identify an innovation that addresses a bottleneck or challenge based on their current experience in health or health care (IP phase 1). This assignment included peers reviewing the video and giving them feedback. The next assignment focused on students identifying their personal views and approach to innovation, including their strengths and weaknesses. The assignment helped students reflect on and identify the leadership skills they need to build, refine, and strengthen to lead innovation in their organization. Another assignment familiarized students with how to search for and review patents already submitted to the US Patent and Trademark Office (phase 2). The students delivered a detailed written summation of their innovation, patent search efforts, and findings.

The students then analyzed the use of models and frameworks in implementing innovation and practice change (phase 4). Applying these tools for implementation creates critical awareness that they must address this common failure point in the innovation pathway. They submitted a paper that described barriers to this innovation change, communication, and strategies to guide the implementation in different contexts and considering the ability to scale and sustain. The last assignment was a group project in which the team developed an innovation to solve a health care problem (all phases). The teams signed nondisclosure agreements focusing on the confidentiality of their projects. They began this project at the start of the course. As they progressed through the modules, their understanding of the foundations of innovation and implementation increased and could be applied to the project.

The final assignment consisted of a term paper and presentation describing the problem, the context of who is experiencing the problem, current treatments and products, competition, market size, revenues, identity regulator and professional organization standards, intellectual property, copyright and trademark, and patents. Next, they outlined and discussed implementation strategies, including a formal decision to release the innovation for production and realistic timelines. A fully developed budget was also submitted (phase 3). The teams presented their projects at the end of the course to potential investors, clients, or users. Classmates participated in the proposals by posing questions or giving feedback in the manner of a hackathon or pitch session. In these assignments, the students learned that innovation is a systematic, continuous, data-driven, and rigorous process of assessment, implementation, and evaluation of tangible innovations.

Outcomes

The digital innovation course was launched in 2020. The course evaluation was designed by a quality committee focusing on the Online Quality Consortium (OQC) framework² and the student's sense of community.³ The OQC focused on course standards, curriculum, and faculty preparation. The sense of community measure was used, given that students expressed the desire to have the university experience despite taking a digital course. There were 5 domains with the evaluation tool that focused on social presence, group participation, faculty presence, technology use, and social media. The anonymous evaluation was an electronic survey using a 4-point scale (4 = strongly agree) administered at the midterm and end of the course. Scores ranged from 2.94 to 3.47. The university evaluation is sent at the end of the term focusing on 8 items (including strong desire to take the class, increased interest, instructor conduct, and workload). The evaluation is rated from 1 to 5 (5 = outstanding). The results from this evaluation ranged from 4.2 to 4.8. Student comments from both evaluations revealed that the professor was engaged, they enjoyed the course modules, the modules were well structured, and the guest speakers provided real-world examples.

Discussion and Conclusion

To realize the Future of Nursing 2020-2030, creating the path to achieve health equity by reducing health disparities focusing on costs, using technology, and maintaining patient and family-focused care innovation will be required.⁴ Schools of nursing can play a critical role in preparing innovative nurse leaders to address pressing issues for the profession of nursing by meeting high-quality and safe patient care. Using the AACN Essentials will guide nursing schools to teach innovation in real time, focusing on new opportunities for improving patient care. Innovation is not a magical process but a deliberate creative series of decisions to solve a problem. This course taught the life cycle of innovation and the nurse leader's role in doing and supporting innovation with their teams and organization—a critical key to success for nursing, health, and health care.

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